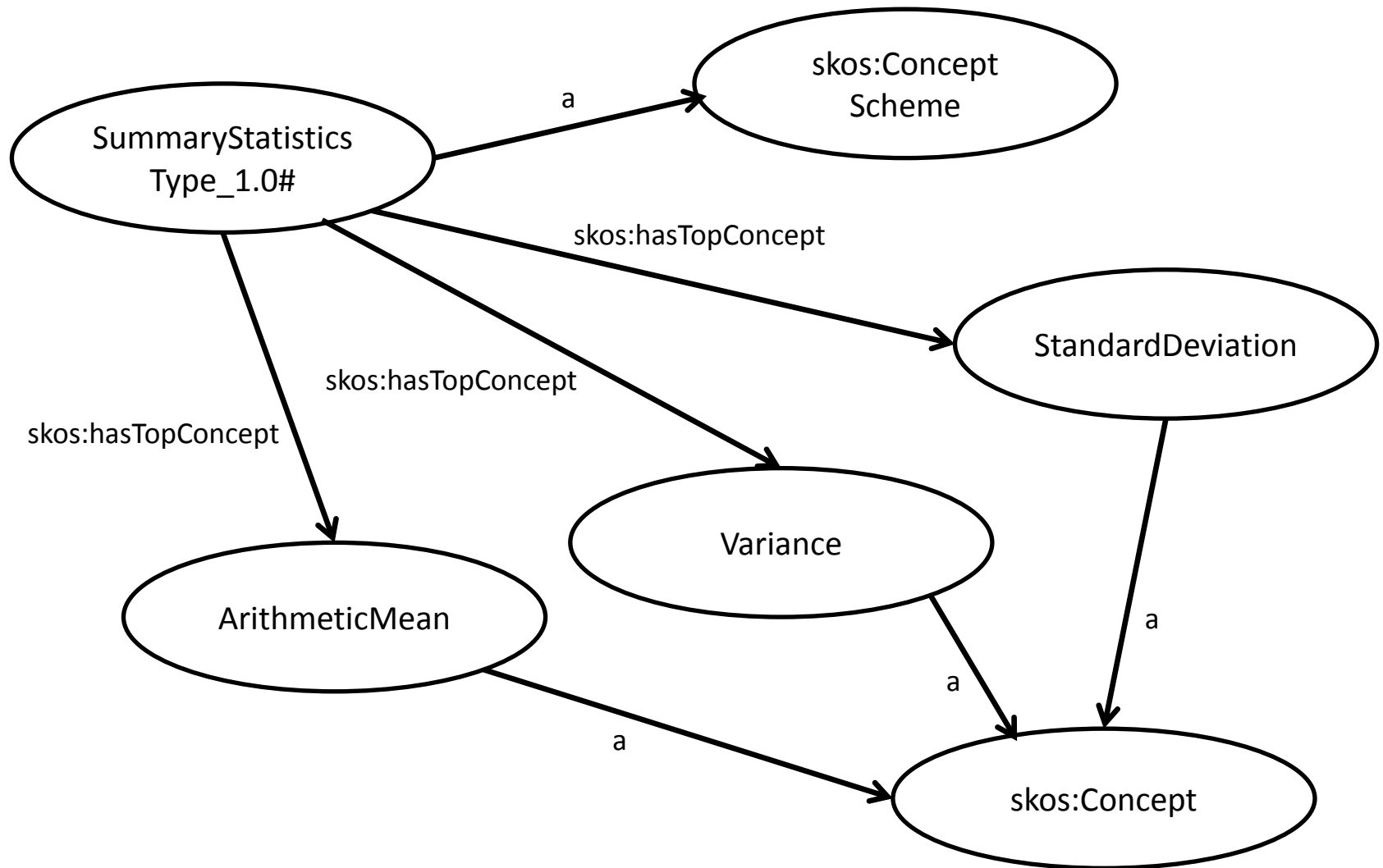




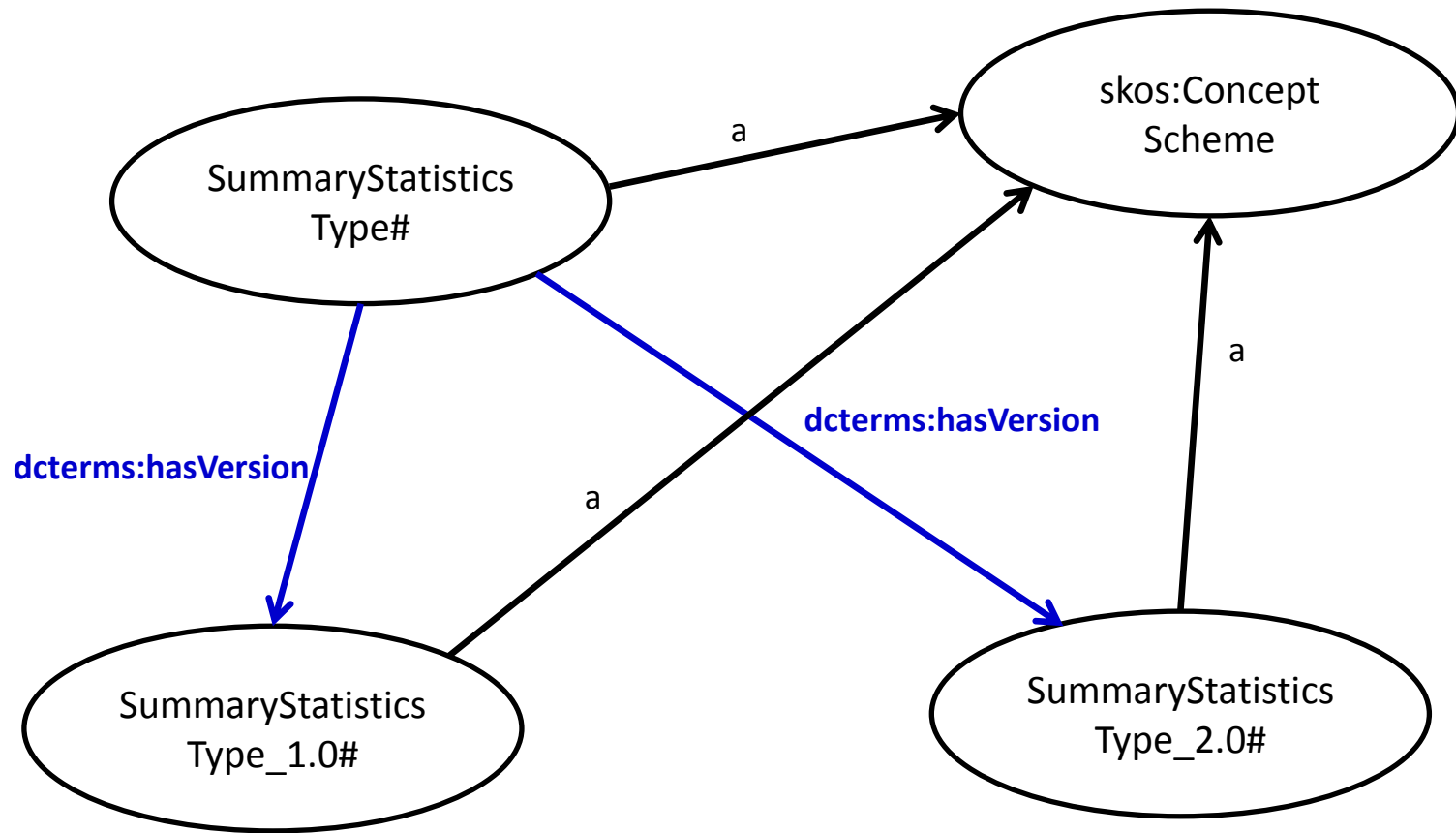
Controlled Vocabularies

Controlled Vocabularies

- Existing DDI-CVs are available in RDF
 - Represented in SKOS format
 - Each CV is a skos:ConceptScheme
 - Each CV entry is a skos:Concept
 - Versioning is considered
- Available at <https://github.com/linked-statistics/DDI-controlled-vocabularies>
- Next step: Review by DDI-CV Working Group



```
<http://rdf-  
vocabulary.ddialliance.org/DDICV/SummaryStatisticType_1.0#Arithm  
eticMean>  
  a skos:Concept ;  
  skos:definition "Mathematical average of a set of values. The  
    mean is calculated by adding up two or more values and  
    dividing the total by their number. In social/political  
    science, it is usually the sum of the measurements divided  
    by the number of subjects, or cases."@en ;  
  skos:inScheme  
    <http://rdf-  
vocabulary.ddialliance.org/DDICV/SummaryStatisticType_1.0#CodeLi  
st> ;  
  skos:notation "ArithmeticMean" ;  
  skos:prefLabel "Arithmetic mean (X)"@en .
```



Versioning

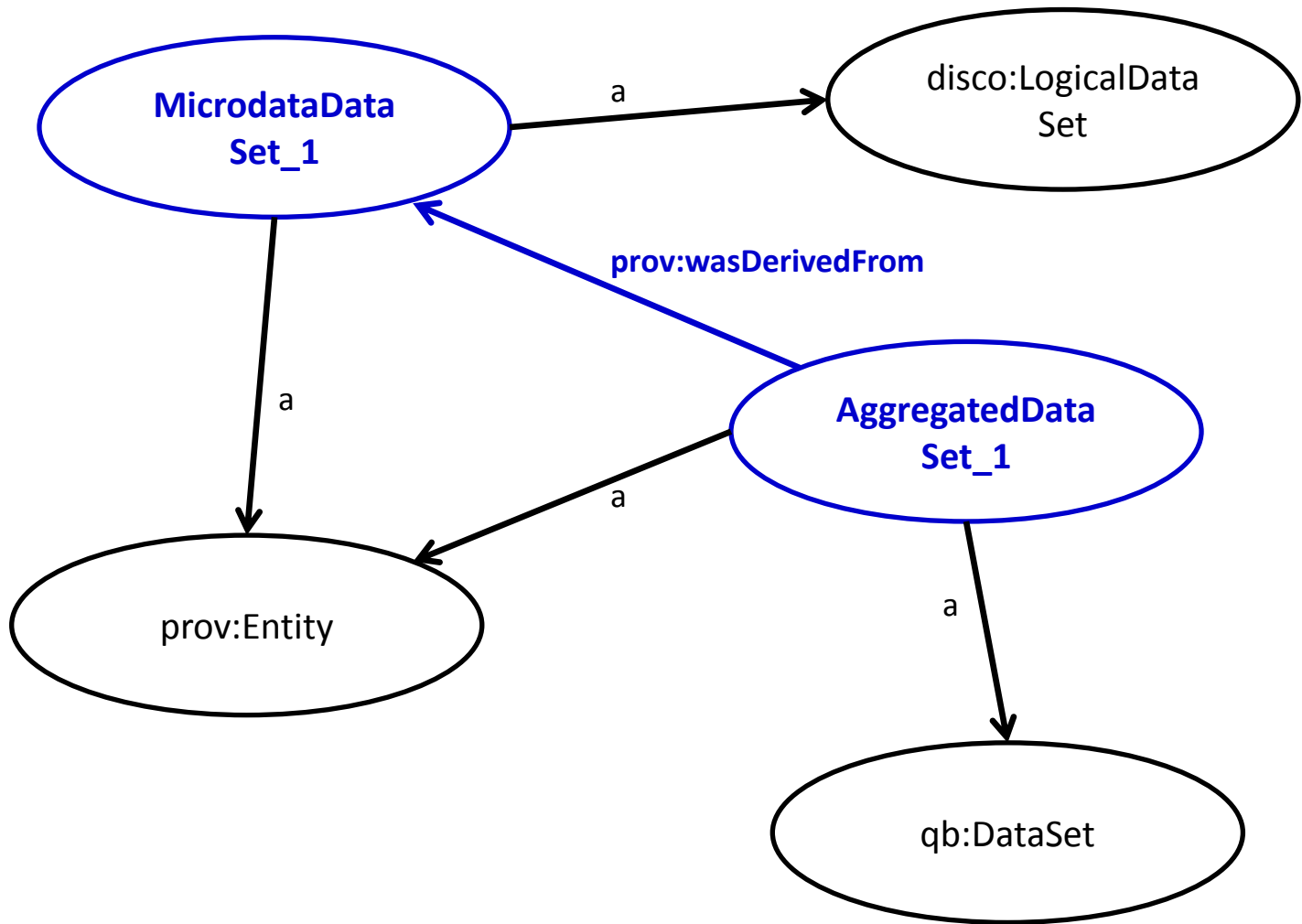
```
<http://rdf-  
vocabulary.ddialliance.org/DDICV/SummaryStatisticType#>  
  a skos:ConceptScheme ;  
  dcterms:title "Base Scheme of Summary Statistic  
    Type"@en ;  
  dcterms:description "Specifies the type of summary  
    statistic. Summary statistics are a single  
    number representation of the characteristics of  
    a set of values."@en ;  
  owl:versionInfo "1.0" ;  
  dcterms:hasVersion  
    <http://rdf-  
vocabulary.ddialliance.org/DDICV/SummaryStatisticType_1.0#  
> ,  
    <http://rdf-  
vocabulary.ddialliance.org/DDICV/SummaryStatisticType_2.0#  
> .
```

Variables

Relationships to other Vocabularies

Relationships to other vocabularies

- Data Cube
 - For representing multidimensional aggregate data
- DCAT
 - For representing collections (catalogs) of research datasets
 - For providing additional information about physical aspects (file size, file formats) of research data files
- PROV-O
 - For representing detailed provenance information, e.g. generation and aggregation of data, versioning information, etc.



Simple Case

```
ddi:AggregatedDataSet_1  
  a prov:Entity ;  
prov:wasDerivedFrom  
  ddi:MicrodataDataSet_1 .
```

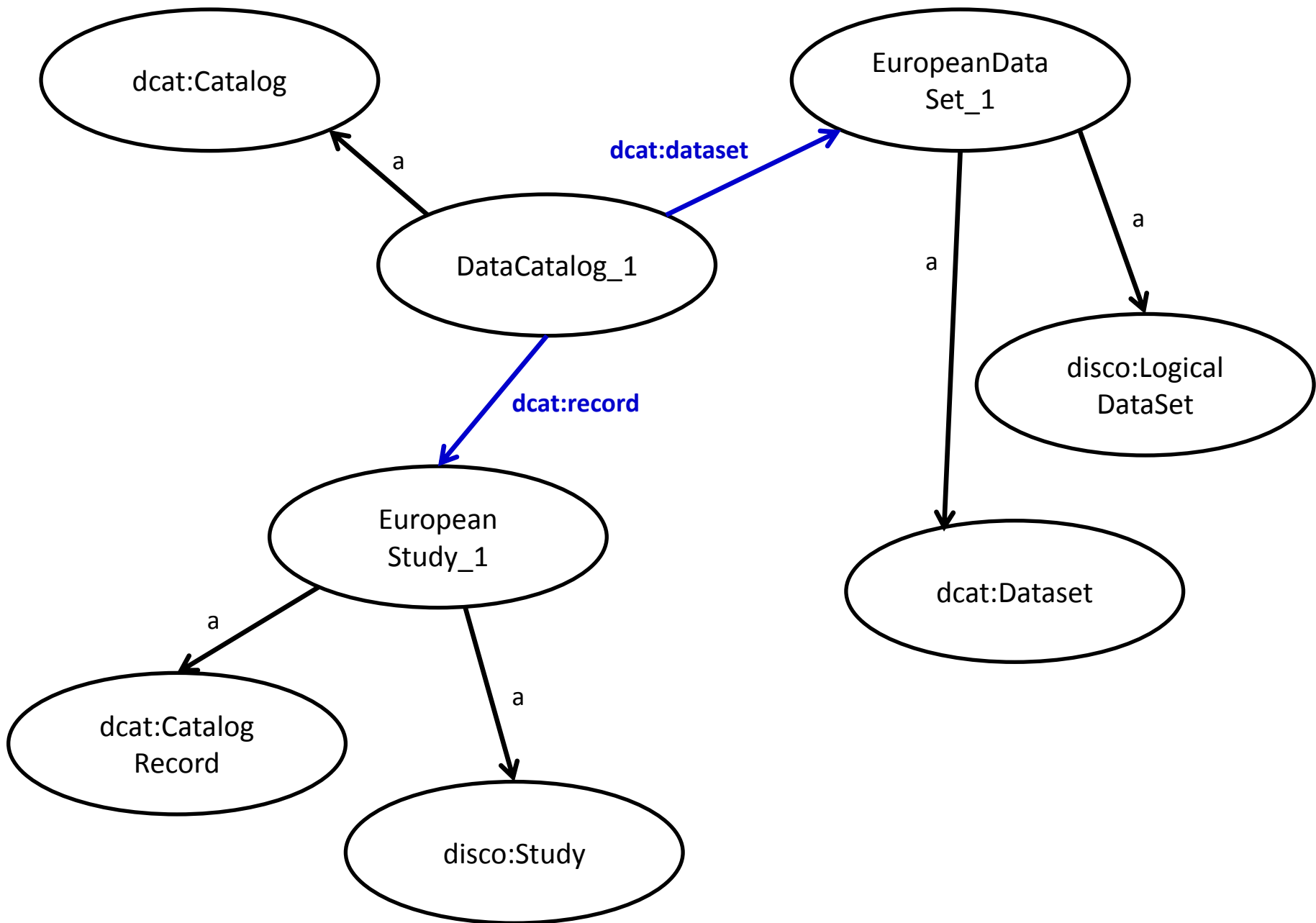
```
ddi:MicrodataDataSet_1  
  a prov:Entity .
```

Complex Case

```
ddi:AggregatedDataSet_2
  a prov:Entity ;
  prov:wasDerivedFrom ddi:MicrodataDataSet_2 ;
  prov:wasGeneratedBy ddi:AggregationActivity ;
  prov:qualifiedDerivation [
    a prov:Derivation ;
    prov:entity ddi:MicrodataDataSet_2 ;
    prov:hadActivity ddi:AggregationActivity
  ] .
```

```
ddi:AggregationActivity
  a prov:Activity .
```

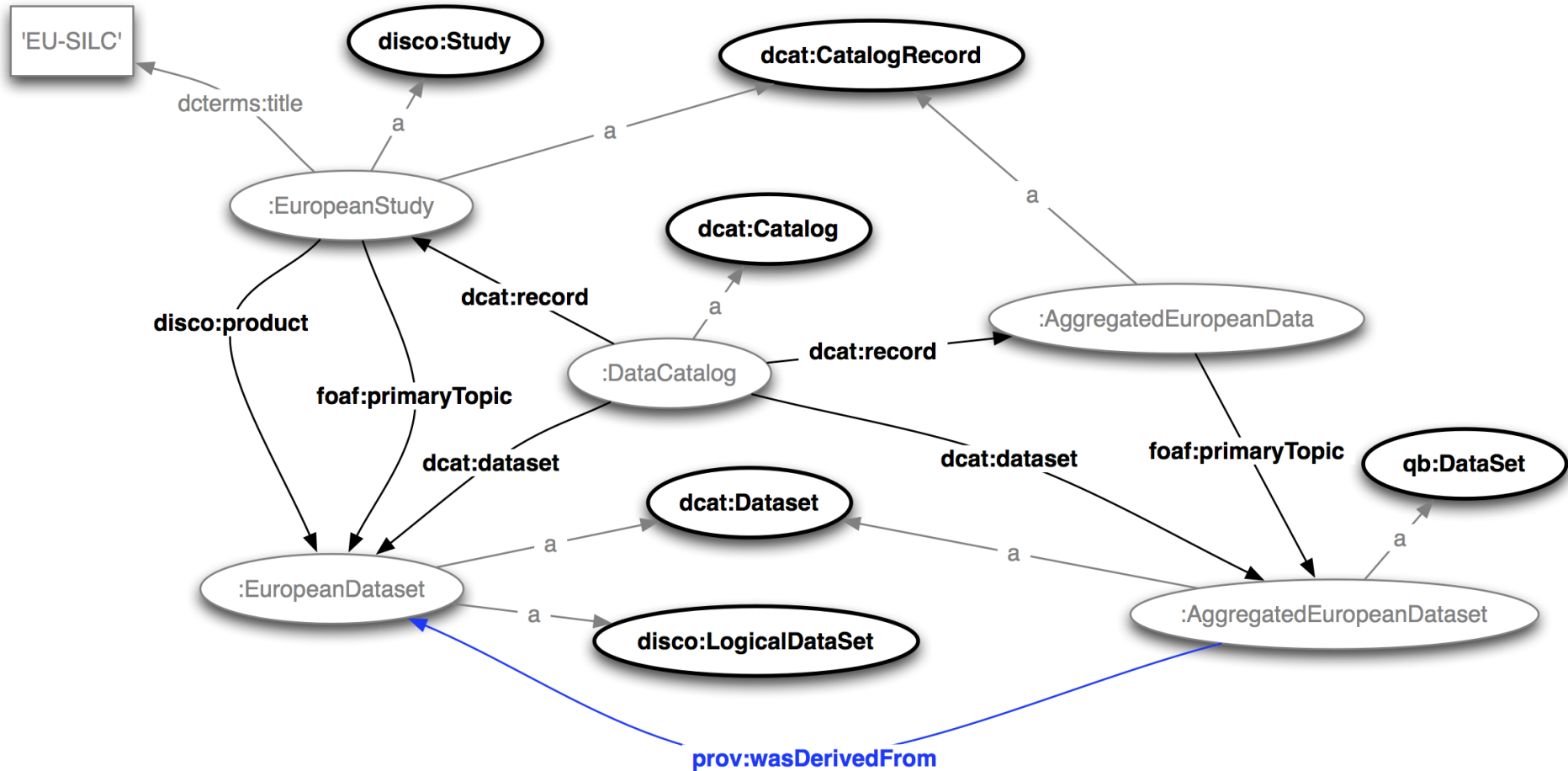
```
ddi:MicrodataDataSet_2
  a prov:Entity .
```



```
ddi:DataCatalog_1
  a dcat:Catalog ;
  dcat:record ddi:EuropeanStudy_1 ;
  dcat:dataset ddi:EuropeanDataSet_1 .
```

```
ddi:EuropeanStudy_1
  a dcat:CatalogRecord, disco:Study ;
  disco:product ddi:EuropeanDataSet_1 .
```

```
ddi:EuropeanDataSet_1
  a dcat:Dataset, disco:LogicalDataSet ;
  dcat:theme ddi:topics/WellBeing ;
  dcat:theme ddi:topics/PoliticalAttitudes ;
  dcat:keyword "Europe"@en ;
  dcat:keyword "Politics"@en .
```



```
ddi:DataCatalog_2
  a dcat:Catalog;
  dcat:record ddi:EuropeanStudy_2 ;
  dcat:record ddi:AggregatedEuropeanData_2 ;
  dcat:dataset ddi:EuropeanDataSet_2 ;
  dcat:dataset ddi:AggregatedEuropeanDataSet_2 .
```

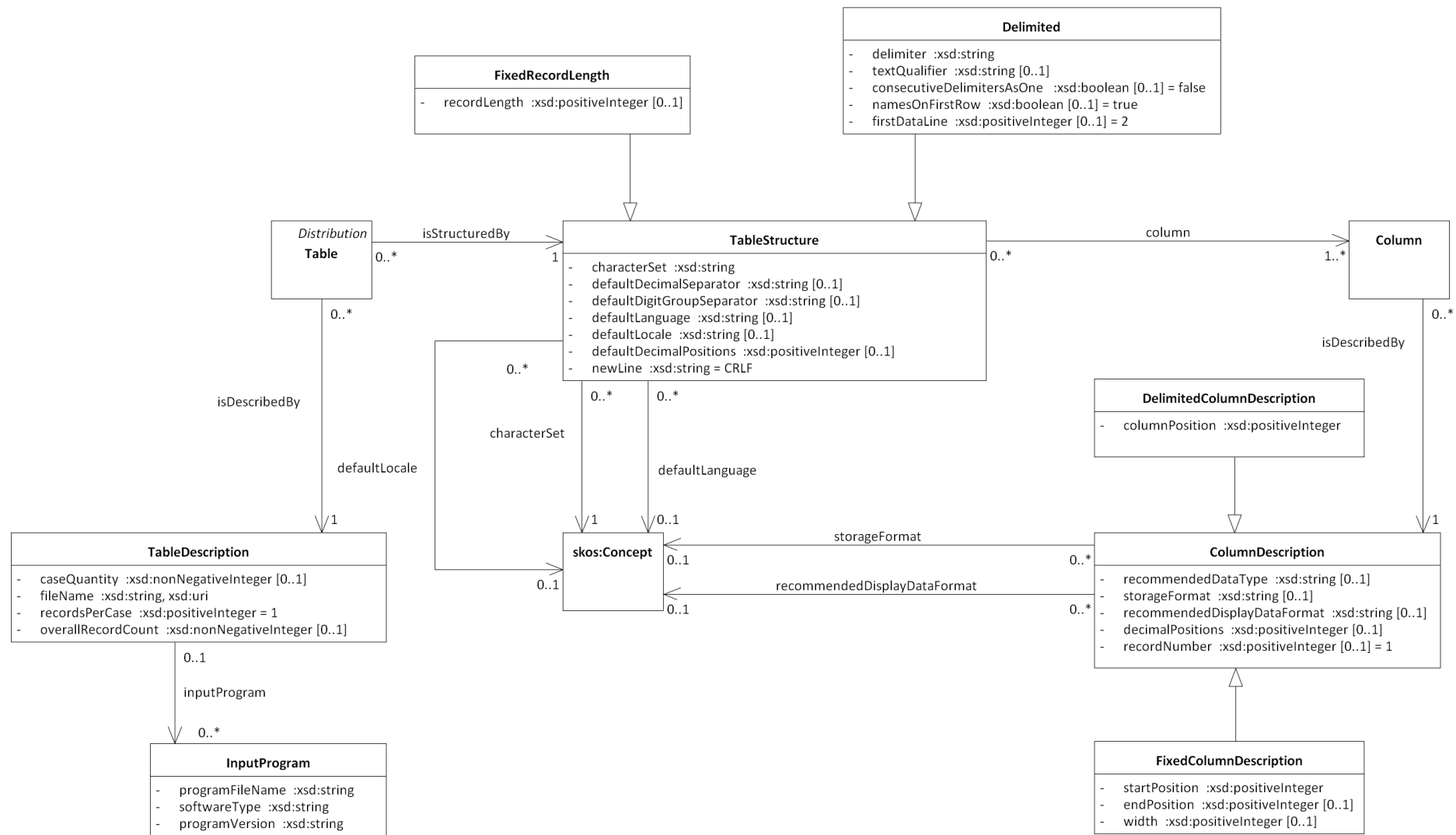
```
ddi:EuropeanStudy_2
  a dcat:CatalogRecord, disco:Study ;
  disco:product ddi:EuropeanDataSet_2 .
```

```
ddi:AggregatedEuropeanData_2
  a dcat:CatalogRecord ;
  foaf:primaryTopic ddi:AggregatedEuropeanDataSet_2.
```

```
ddi:EuropeanDataSet_2
  a dcat:Dataset, disco:LogicalDataSet .
```

```
ddi:AggregatedEuropeanDataSet_2
  a dcat:Dataset, qb:DataSet ;
  prov:wasDerivedFrom ddi:EuropeanStudy_2 .
```

PHDD



Mapping DDI-XML to Disco

Mapping DDI-XML to Disco

- Mappings only between Disco and DDI 3.1 of DDI-L in order to avoid inconsistencies
 - existing mapping documents between DDI 3.1 and other DDI versions (like DDI 3.2 and DDI 2.1) can be reused
- Availability
 - Google Doc with mapping tables as basis for automatic generation
 - Turtle file containing all mappings
 - Mapping tables in HTML specification of Disco
- Mapping is still ongoing work

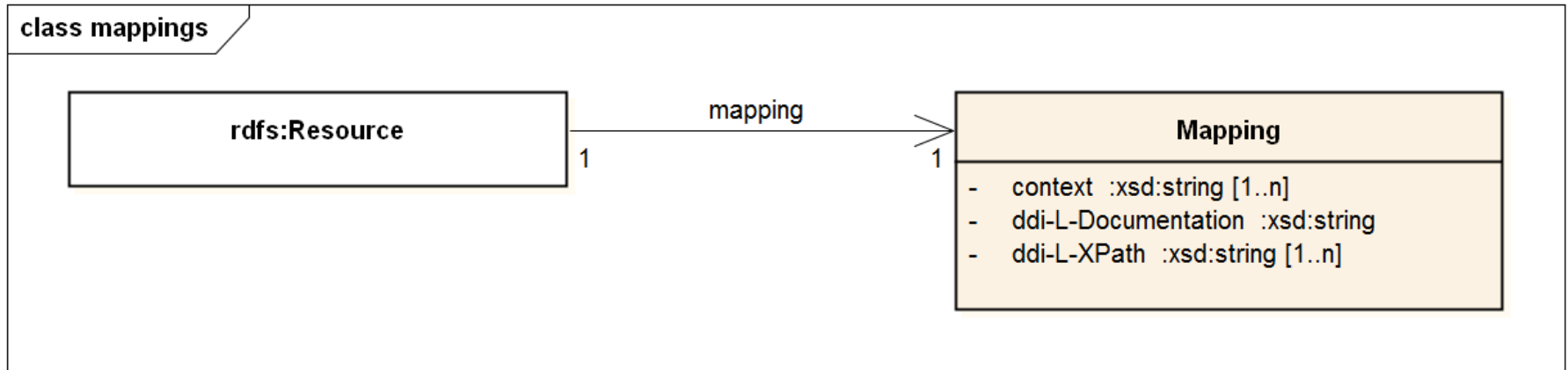
XSLT for existing DDI-XML

- XSLTs for converting any XML output of DDI-C and DDI-L are available at <https://github.com/linked-statistics/DDI-RDF-tools>
- Different XSLT for DDI-C and DDI-L

Bidirectional Mappings

- Only between Disco and DDI-L
 - DDI-L $\rightsquigarrow$ Disco: straight-forward mapping for all items used in Disco
 - Disco $\rightsquigarrow$ DDI-L: straight-forward mapping for all items in the disco namespace.
- Only standard XPath expression is defined as mapping
- Context:
 - Items from other vocabularies - used in Disco - need a context; then there could be a clear mapping path.
 - Context information necessary for mappings, e.g., skos:notation can be mapped to variable labels and to codes.
 - Context information is either a SPARQL query or an informal description as plain literal.

Mapping Representation



- Mapping ontology available containing all mapping triples
- generated automatically out of the official mapping document

Mapping Representation

```
skos:notation
  a rdfs:Class, owl:Class ;
disco:mapping [
  a disco:Mapping ;
  disco:ddi-L-Xpath
    "//l:Variable/l:VariableName" ;
  disco:ddi-L-Documentation
    "http://www.ddialliance.org/Specification/DDI-
    Lifecycle/3.1/XMLSchema/FieldLevelDocumentatio
    n/logicalproduct_xsd/elements/V
    ariable.html"
  disco:context
    "skos:notation represents variable label" ;
  disco:context
    "SELECT ?notation WHERE { ?notation
rdfs:domain ?variable. ?variable a
disco:Variable. }" ]
```

DDI 4

**Let's Disco
Now!**



Acknowledgements

26 experts from the statistical community and the Linked Data community coming from 12 different countries contributed to this work. They were participating in the events mentioned below.

- [1st workshop](#) on 'Semantic Statistics for Social, Behavioural, and Economic Sciences: Leveraging the DDI Model for the Linked Data Web' at Schloss Dagstuhl - Leibniz Center for Informatics, Germany in September 2011
- Working meeting in the course of the 3rd Annual European DDI Users Group Meeting ([EDDI11](#)) in Gothenburg, Sweden in December 2011
- [2nd workshop](#) on 'Semantic Statistics for Social, Behavioural, and Economic Sciences: Leveraging the DDI Model for the Linked Data Web' at Schloss Dagstuhl - Leibniz Center for Informatics, Germany in October 2012
- Working meeting at GESIS - Leibniz Institute for the Social Sciences in Mannheim, Germany in February 2013